# Vowels

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```
source(here::here("_extensions/mcanouil/typst-render/
_resources/typst_define.R"))
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(ggplot2)
library(colorspace)
library(tidynorm)
library(ggdensity)
library(scales)
```



```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
library(geomtextpath)
library(ggdist)
```

```
reul_tri <- function(theta, n=3){
  tau <- 2 * pi
  f = floor(
    ((n * (theta-pi))/tau) + 0.5
  )
  a = cos(theta - (tau/n)*f)

  a + sqrt(1 + (2*cos(pi/n)) + (a^2))
}
```

## Vowels

What is a vowel?

- Not a consonant.

## Vowels

Clues in the name

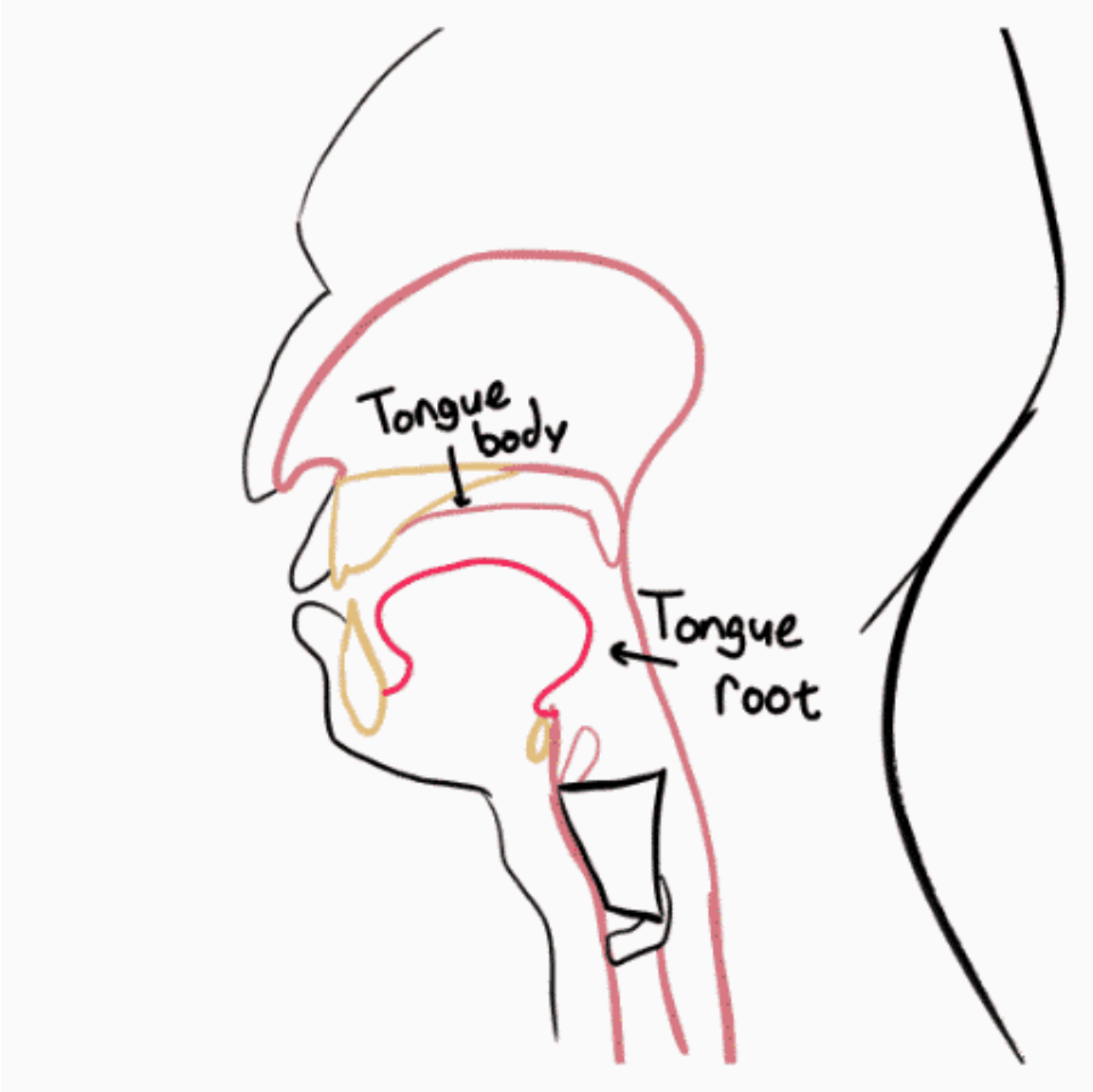
| | English | Dutch |
|---|---|---|
| V | (sonant?) | klinker |
| C | consonant | medeklinker |

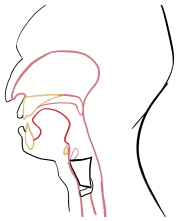
## Vowels

In general, a more open, unconstricted, artiulation compared to consonants.

## Vowels - Articulatory Dimensions

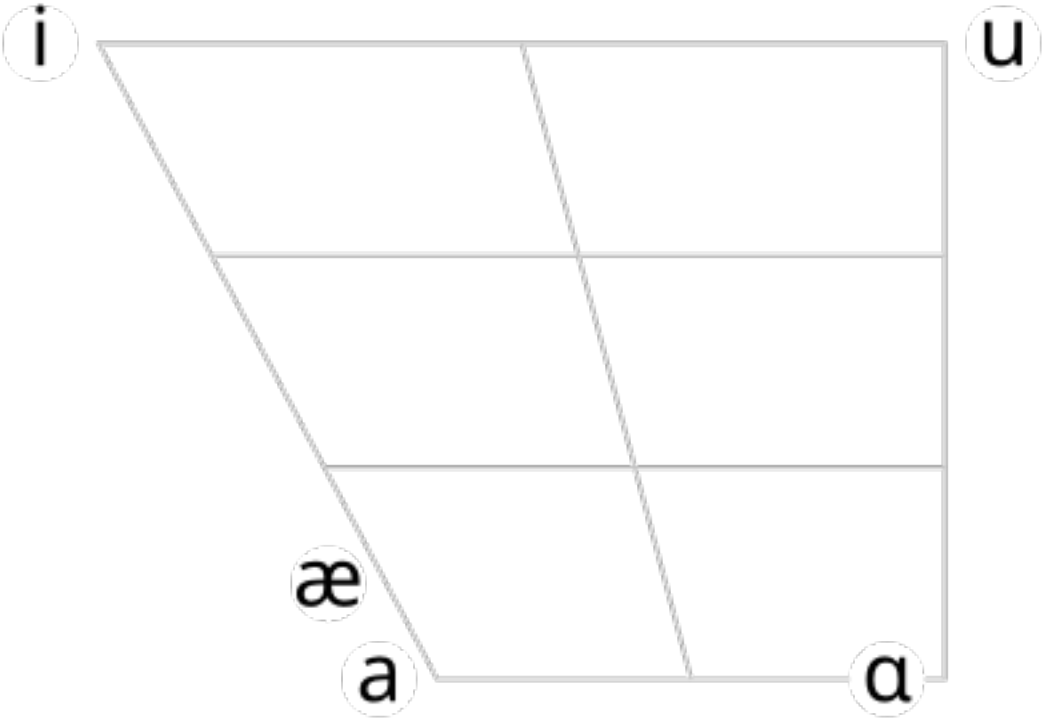
Tongue body and Tongue Root




| | Front | Back |
|---|---|---|
| **High** |  |  |

| | Front | Back |
|---|---|---|
| **Low** |  |  |

## Vowels - Dimensions




## Vowels - Transcription

Vowels *fundamentally* exist in a continuous phonetic space, making the transcription problem especially tricky.

- It can be tricky putting a categorical label on a continuous space.
- Many “conventional” transcriptions use symbols that aren’t technically accurate.

## Vowels - Transcription

# “goose”

# gu:s

?

## Vowels - Transcription

```
anae <- read_rds("~/OneDrive - University of Kentucky/Corpora/ANAE/rds_data/anaeUNNORM.rds")
```

```
anae |>
  norm_nearey(
    .by = TS,
    F1:F2
  ) ->
  anae_norm
```

```
Normalization info
• normalized with `tidynorm::norm_nearey()`
• normalized `F1` and `F2`
• normalized values in `F1_lm` and `F2_lm`
• grouped by `TS`
• within formant: FALSE
• Transformation prior to normalization: log
• (.formant - mean(.formant, na.rm = T))/(1)
```

```
anae_norm |>
  ggplot(
    aes(F2_lm, F1_lm)
  ) +
  stat_hdr(
    probs = 0.95,
    color = "black",
    fill = "grey90",
    show.legend = F
  ) +
  scale_x_reverse() +
  scale_y_reverse() +
  theme_no_x() +
  theme_no_y() +
  coord_fixed() ->
  anae_base

make_means <- function(df){
```

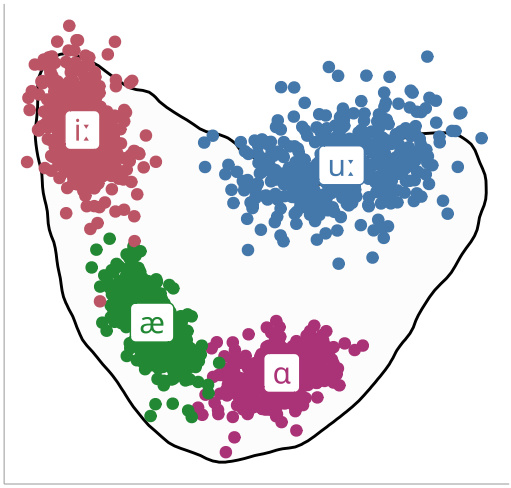
```
  df |>
    summarise(
      .by = matches("[^xy]"),
      across(c(x,y), mean)
    )
}
```

```
anae_norm |>
 summarise(
    .by = c(TS, VClass),
    across(F1_lm:F2_lm, mean)
  ) ->
  speaker_means

speaker_means |>
  filter(VClass %in% c("iy", "uw", "ae", "o")) |>
  mutate(
    vowel = recode_values(
      VClass,
      "iy" ~ "iː",
      "ae" ~ "æ",
      "uw" ~ "uː",
      "o" ~ "ɑ"
    )
  ) |>
  mutate(
    vowel = fct_reorder(vowel, F1_lm)
  ) ->
  vpoints
```

```
anae_base +
  geom_point(
    data = vpoints,
    aes(color = vowel),
    show.legend = F
  ) +
  geom_label(
    data = vpoints,
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
```

```
  show.legend = F
)
```

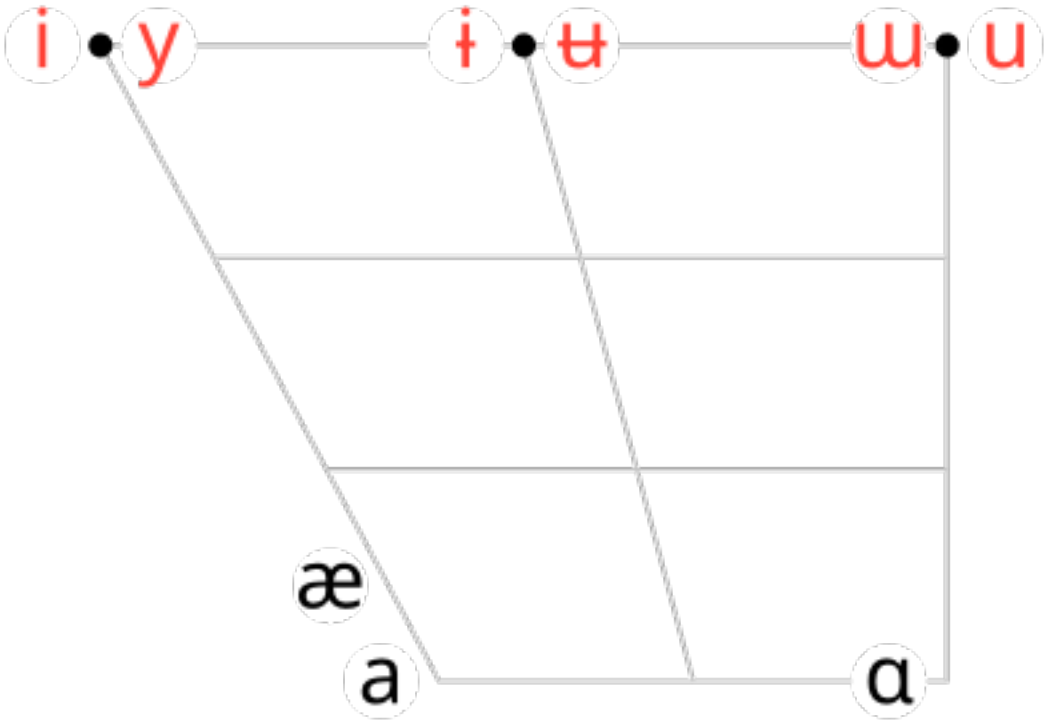



Data from the Atlas of North American English

1 point = 1 speaker

## Vowels - Transcription

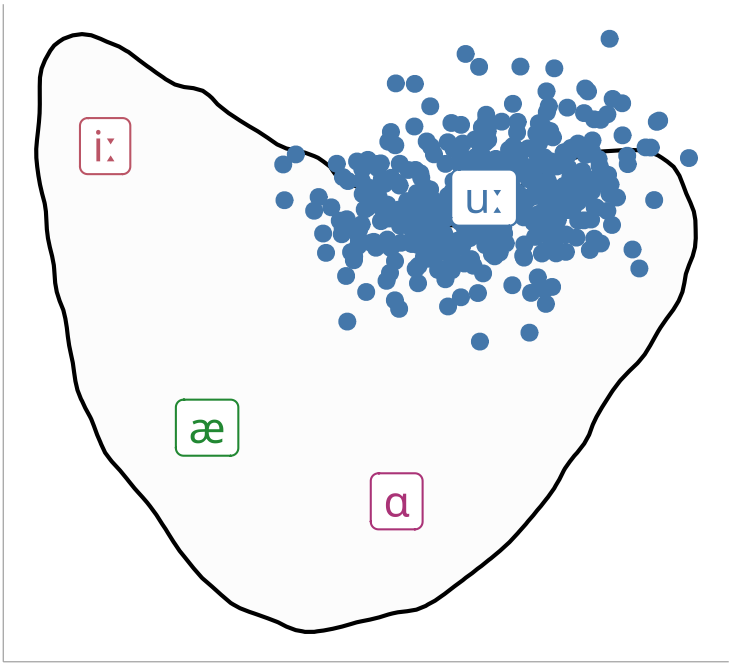
### High Vowels





## Vowel - Transcription

```
anae_base +
  geom_label(
    data = vpoints,
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F
  ) +
  geom_point(
    data = vpoints |> filter(VClass == "uw"),
    aes(color = vowel),
    show.legend = F
  ) +
  geom_label(
    data = vpoints |>  filter(VClass == "uw"),
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F
  )
```




Vowels - Transcription

# gɨs?

# gʉs?

Vowels - Transcription

# “thought”

# θɔt

?

Vowels - Transcription

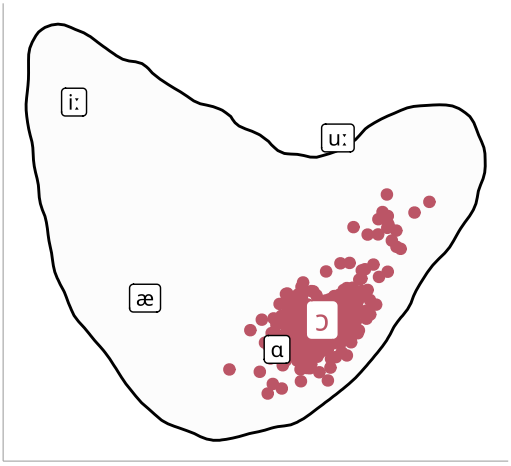
```
anae_norm |>
  filter(VClass %in% c("oh")) |>
  summarise(
    .by = c(TS, VClass),
    across(F1_lm:F2_lm, mean)
```

```
  ) |>
  mutate(
    vowel = case_match(
      VClass,
      "oh" ~ "ɔ"
    )
  ) |>
  mutate(
    vowel = fct_reorder(vowel, F1_lm)
  ) ->
  oh
anae_base +
  geom_point(
    data = oh,
    aes(color = vowel),
    show.legend = F
  ) +
  geom_label(
    data = oh,
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F
  ) +
  geom_label(
    data = vpoints,
    aes(label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F,
    size = 4
  ) +
  theme(
    text = element_text(family = "Voces")
  )
```

```
Warning: There was 1 warning in `mutate()`.
ℹ In argument: `vowel = case_match(VClass, "oh" ~ "ɔ")`.
Caused by warning:
! `case_match()` was deprecated in dplyr 1.2.0.
ℹ Please use `recode_values()` instead.
```

Data from the Atlas of North American English

1 point = 1 speaker




## Vowels - Transcription

Back Vowels

**[typst-render] Compilation failed for 'typst-block-3'.**

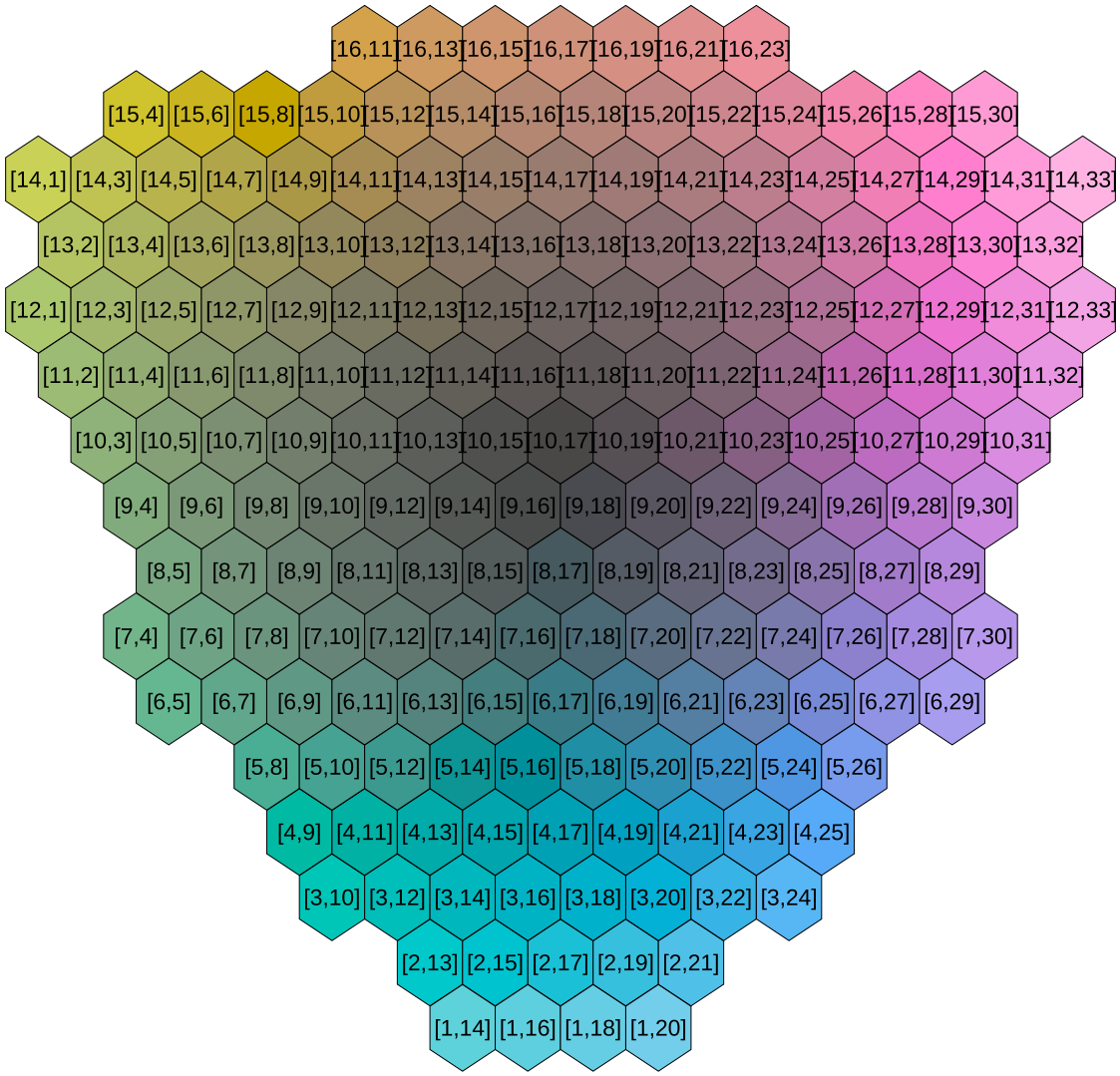
## Vowels - Transcription

θot?

θɔt?

θɒt?

## Vowels?



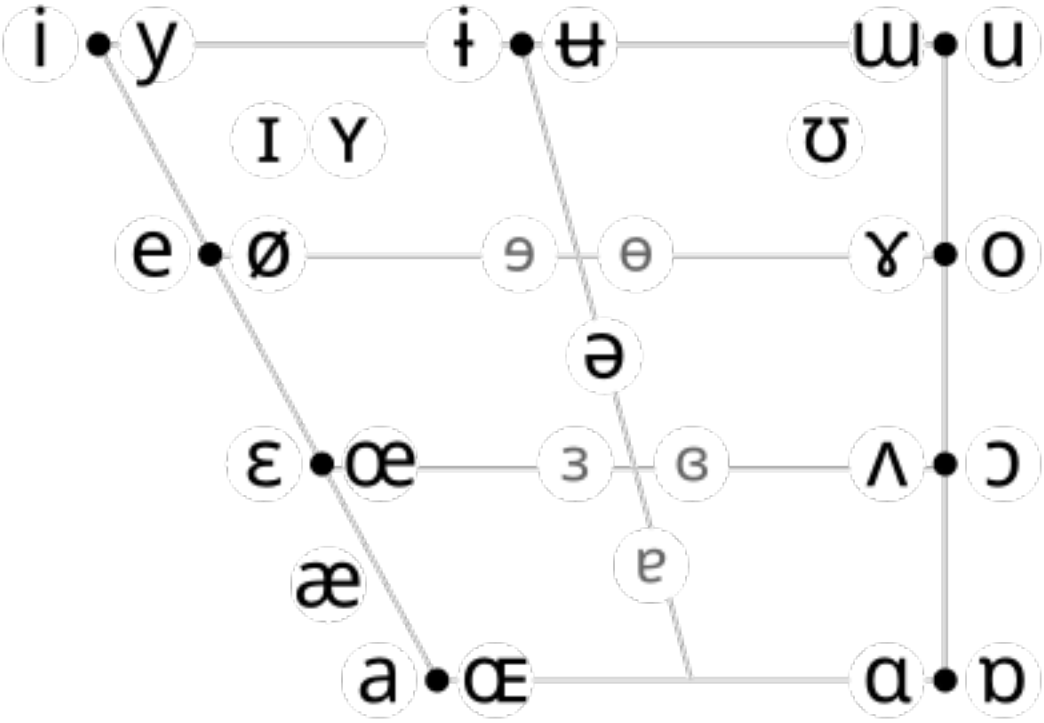

## Vowels - Transcription

If *all* we have is paper and ink to describe phonetic detail, we should try to use the closest IPA character.

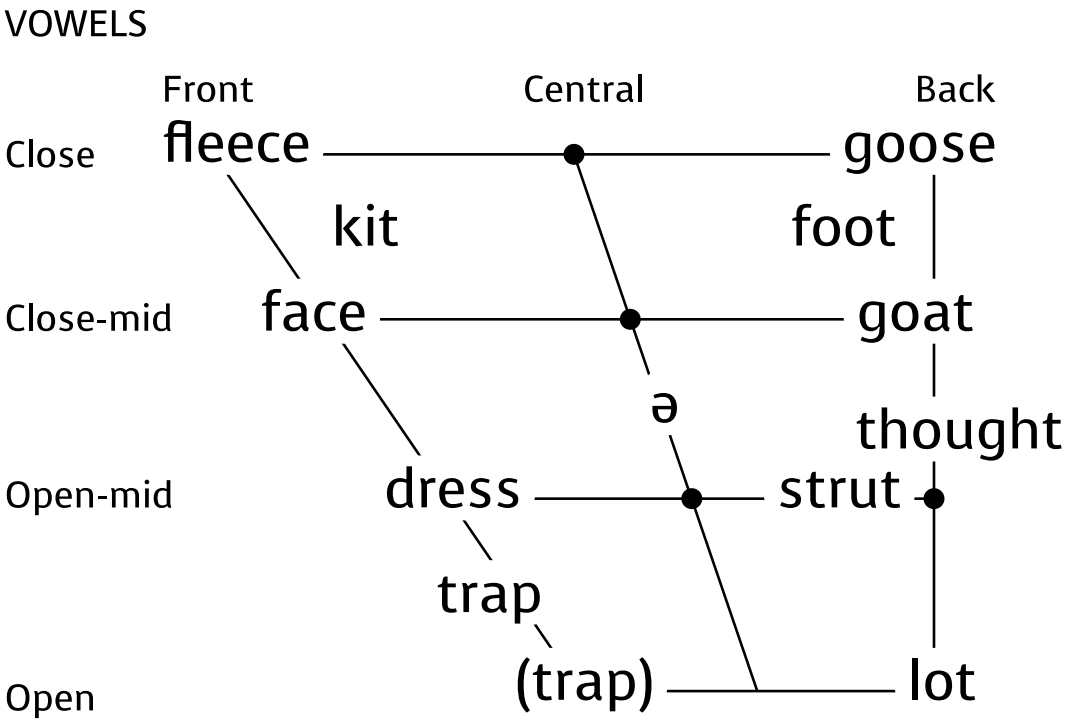
Otherwise, we'll use symbols that best characterize lexical sets.

[u:] ∈ {goose, coop, food}

Vowels

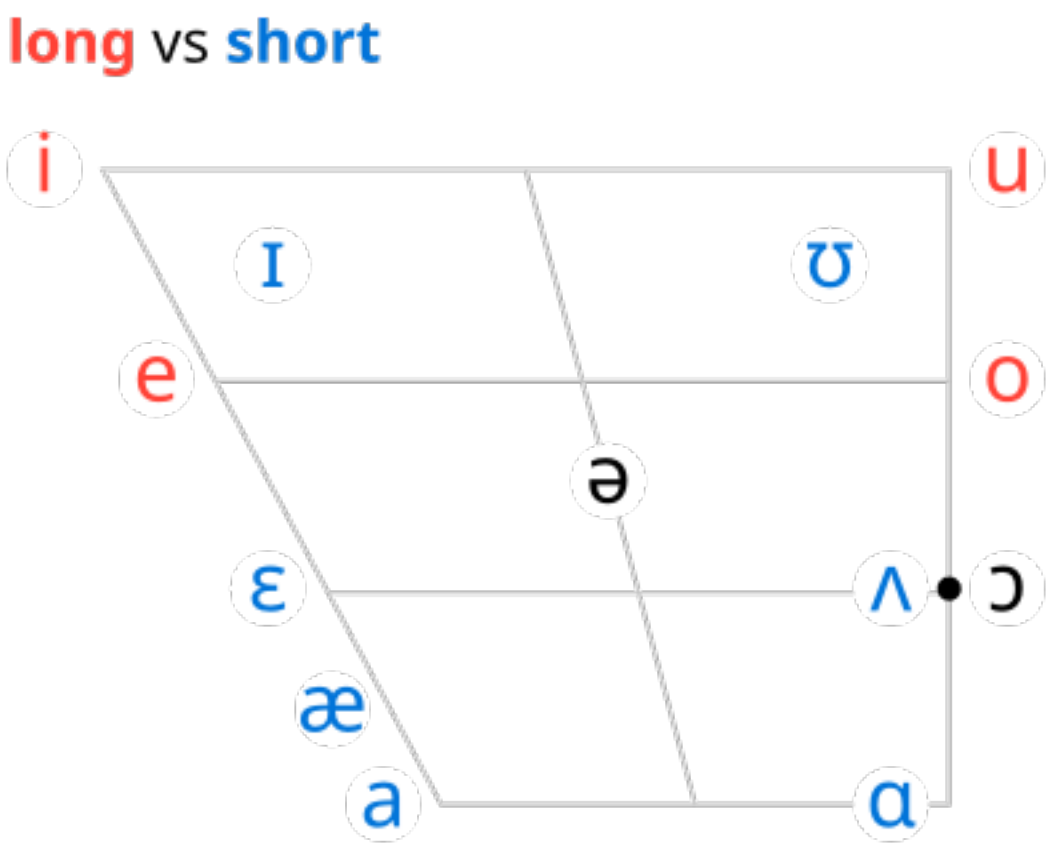


Vowels - English Keywords




Vowels - Length

In English




Vowels - Length

Length is transcribed with a ː symbol (which is different from :)!

iː

uː

Vowels - Length

In English, the mid long vowels tend to be diphthongized.

Vowel - Length

The mid long vowels in English tend to be diphthongized

```
jean_lp <- read_csv("~/OneDrive - University of Kentucky/Corpora/PNC/derived_data/fave_results/PH06-2-1-AB-Jean_logparam.csv")
```

```
Rows: 34635 Columns: 22
──                                                                    Column
specification ──────────────────────────────────────────────────────────────
Delimiter: ","
chr (12): file_name, id, group, label, word, pre_word, fol_word, pre_seg, fo...
dbl (10): param, F1, F2, F3, error, max_formant, speaker_num, optimized, str...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
jean_lp |>
  filter(param == 0) |>
  mutate(across(F1:F2, \(x)x*sqrt(2))) |>
  ggplot(
    aes(F2, F1)
  )+
  stat_hdr(
    color = "black",
    fill = "grey80",
    probs = 0.95,
    show.legend = F
  ) +
  scale_x_reverse() +
  scale_y_reverse() +
  coord_fixed() +
  theme_no_x() +
  theme_no_y() ->
  jean_base
```

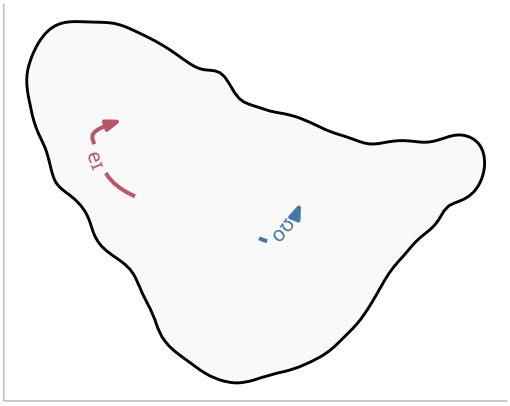
```
jean_lp |>
  filter(
    label %in% c("ey", "ow"),
    fol_seg != "l"
  ) |>
  summarise(
    .by = c(param, label),
    across(F1:F2, mean)
  ) |>
  reframe_with_idct(
    .param_col = param,
    .token_id_col = label,
    F1:F2
  ) |>
```



```
  mutate(
    label = case_match(
      label,
      "ey" ~ "eɪ",
      "ow" ~ "oʊ"
    )
  ) ->
  jean_ey_ow
jean_base +
  geom_textpath(
    data = jean_ey_ow,
    aes(
      color = label,
      label = label
    ),
    linewidth = 1,
    arrow = arrow(
      type = "closed",
      length = unit(0.25, "cm"),
      angle = 20
    ),
    family = "Noto Sans",
    show.legend = F
  )
```

```
Warning in reframe_with_idct(summarise(filter(jean_lp,
label %in% c("ey", : There is was no
grouping provided.
ℹ You may want to provide `.by` with
a speaker id column.
```

Average [eɪ] and [oʊ] from one speaker from Philadelphia.




## Vowels - Diphthongs

```
jean_lp |>
  filter(
    label %in% c("ay", "aw", "oy")
  ) |>
  summarise(
    .by = c(param, label),
    across(F1:F2, mean)
  ) |>
  reframe_with_idct(
    .param_col = param,
    .token_id_col = label,
    F1:F2
  ) |>
  mutate(
    label = case_match(
      label,
      "ay" ~ "aɪ",
      "aw" ~ "aʊ",
      "oy" ~ "ɔɪ"
    )
```

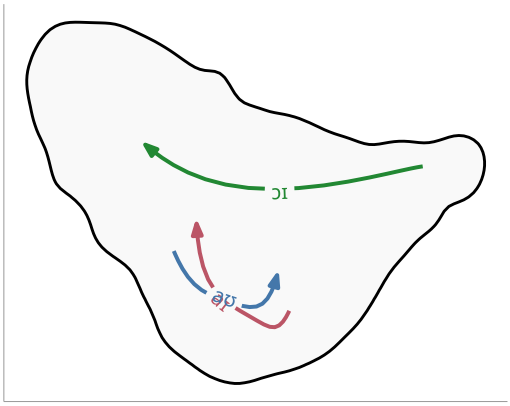
```
  )->
  jean_diph
```

```
Warning in reframe_with_idct(summarise(filter(jean_lp, label %in% c("ay", : There
is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```

```

```
jean_base+
  geom_textpath(
    data = jean_diph,
    aes(
      color = label,
      label = label
    ),
    linewidth = 1,
    arrow = arrow(
      type = "closed",
      length = unit(0.25, "cm"),
      angle = 20
    ),
    family = "Noto Sans",
    show.legend = F
  )
```

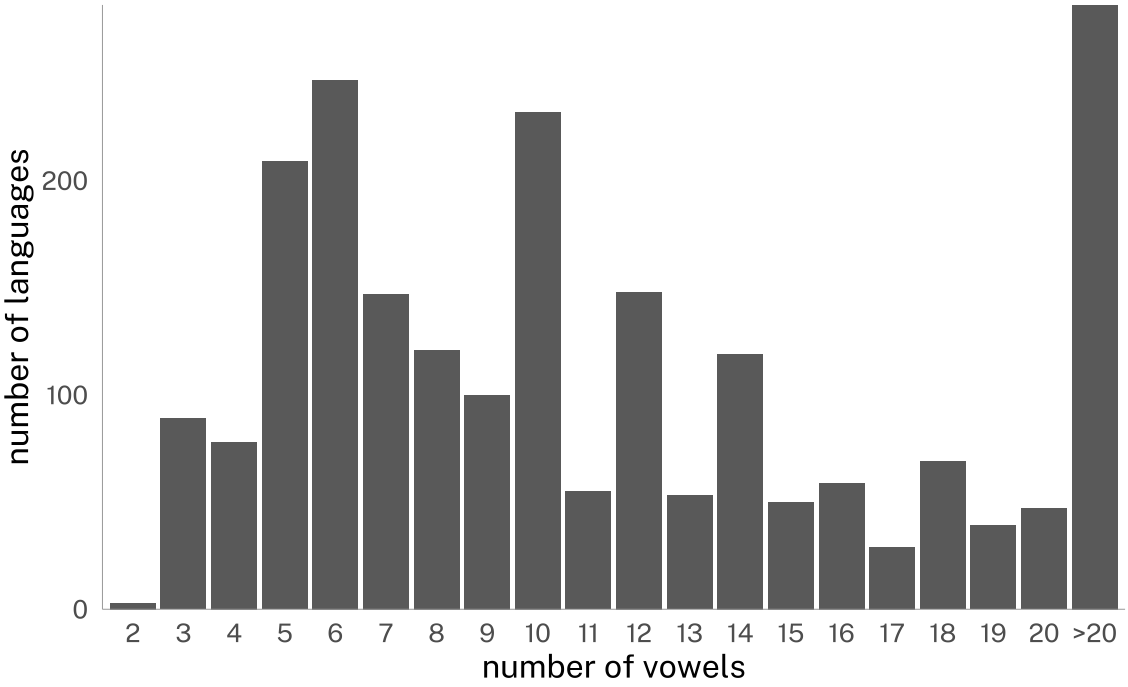



Three diphthongs from a speaker from Philadelphia



# Across the world

## Vowel Inventories




## Inventories

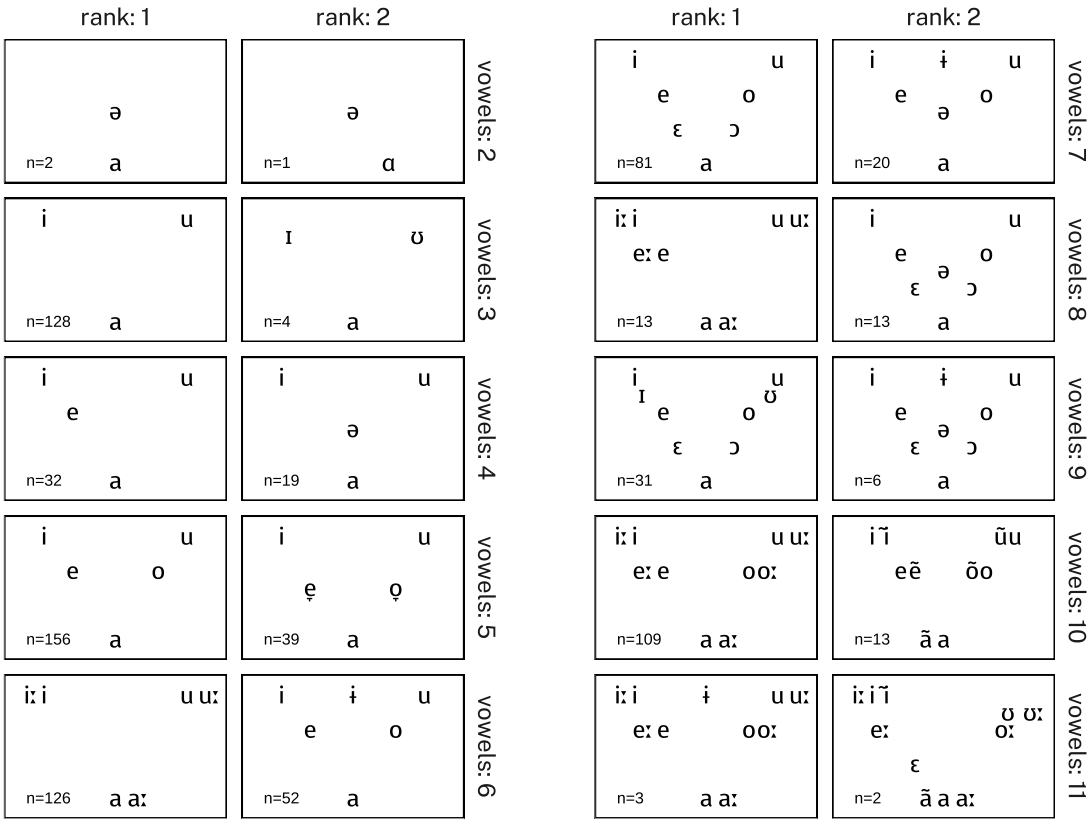
```
phoible |>
  filter(
    SegmentClass == "vowel"
  ) |>
  mutate(
    .by = c(Glottocode, SpecificDialect),
    vowels = n()
  ) |>
  count(
    vowels, Phoneme
  ) |>
  arrange(
    desc(n)
  ) |>
  slice(
    .by = vowels,
    1:first(vowels)
  ) |>
```

```
  arrange(vowels) ->
  vowel_counts
```

```
tribble(
  ~Phoneme, ~y, ~x,
  "ɑ", 0, 1,
  "a", 0, 0,
  "aː", 0, 0.6,
  "ã", 0, -0.5,
  "æ", 0, -0.7,
  "e", 2, -1.2,
  "eː", 2, -1.8,
  "ẽ", 2, -0.8,
  "ø", 2, -0.8,
  "i", 3, -2,
  "iː", 3, -2.4,
  "y", 3, -1.6,
  "ĩ", 3, -1.6,
  "o", 2, 1.2,
  "oː", 2, 1.7,
  "õ",2, 0.8,
  "ɤ", 2, 0.8,
  "u", 3, 2,
  "uː", 3, 2.6,
  "ũ", 3, 1.6,
  "ɯ", 3, 1.6,
  "ɔ", 1, 0.8,
  "ə", 1.5, 0,
  "ɛ", 1, -0.8,
  "œ", 1, -0.4,
  "ɪ", 2.5, -1.8,
  "ʏ", 2.5, -1.4,
  "ʊ", 2.5, 1.8,
  "ʊː", 2.5, 2.5,
  "e̞", 1.5, -1.2,
  "o̞", 1.5, 1.2,
  "ɨ", 3, 0,

  # "ɯ", "ɪ", "ʊ",
) ->
  vowel_coords
```

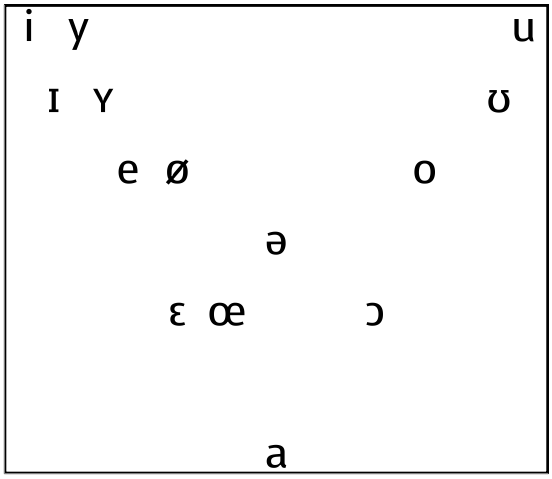
```
Joining with `by = join_by(Phoneme)`
```




## Inventories

```
Joining with `by = join_by(Phoneme)`
```

Muyang




ngwe

